**Listing 8.1 `document_vector_similarity` (transform to vector)**

```
import plotly.graph_objects as go
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [      #1
    "The sky is blue and beautiful.",
    "Love this blue and beautiful sky!",
    "The quick brown fox jumps over the lazy dog.",
    "A king's breakfast has sausages, ham, bacon, eggs, toast, and beans",
    "I love green eggs, ham, sausages and bacon!",
    "The brown fox is quick and the blue dog is lazy!",
    "The sky is very blue and the sky is very beautiful today",
    "The dog is lazy but the brown fox is quick!"
]

vectorizer = TfidfVectorizer()     #2
X = vectorizer.fit_transform(documents)      #3
```

**#1 Samples of documents**
**#2 Vectorization using TF–IDF**
**#3 Vectorize the documents.**

Let's break down TF–IDF into its two components using the sample sentence, "The sky is blue and beautiful," and focusing on the word *blue*.

## *TERM FREQUENCY (TF)*

*Term Frequency* measures how frequently a term occurs in a document. Because we're considering only a single document (our sample sentence), the simplest form of the TF for *blue* can be calculated as the number of times *blue* appears in the document divided by the total number of words in the document. Let's calculate it:

Number of times *blue* appears in the document: 1

Total number of words in the document: 6

TF = 1 ÷ 6TF = .16

## *INVERSE DOCUMENT FREQUENCY (IDF)*

*Inverse Document Frequency* measures how important a term is within the entire corpus. It's calculated by dividing the total number of documents by